The Blueprint of Life

Every cell in your body has the same "blueprint" or the same DNA. Like the blueprints of a house tell the builders how to construct a house, the DNA "blueprint" tells the cell how to build the organism. Yet, how can a heart be so different from a brain if all the cells contain the same instructions? Although much work remains in genetics, it has become apparent that a cell has the ability to turn off most genes and only work with the genes necessary to do a job.

To build the body, the code carried on messenger RNA travels to the ribosomes where three bases within the code (called a **codon**) specify a single **amino acid**. A long mRNA chain could contain thousands of these amino acids. When linked together, the amino acids form **proteins**, which make up many structures in the body.

In this section, colour the bases and RNA chain the same as you did previously. Colour the amino acids red. ☐

DNA Replication

Each time a new cell is made, the cell must receive an exact copy of the parent cell DNA. The new cells then receive the instructions and information needed to function. The process of copying DNA is called **replication**. Replication occurs in a unique way – instead of copying a complete new strand of DNA, the process “saves” or conserves one of the original strands. For this reason, replication is called **semi-conservative**. When the DNA is ready to copy, the molecule “unzips” itself and new nucleotides are added to each side.

The image showing replication is similar to the DNA and mRNA colouring. Note the nucleotides are shown as 2 parts – the sugar and phosphate (colour blue) and one of the four bases identified by shape, colour the same as you did on the other model. ☐

Questions:

1. Why is the nucleus called the "control center" of the cell?

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2. What is a gene?

_____

3. Where in the cell are chromosomes located?

_____

4. DNA can be found in what **organelles**?

_____

5. What two scientists established the structure of DNA?

_____

6. Replication is called “semi-conservative” because half of the original strand is

_____

7. What are the sides of the DNA ladder made of?

_____

8. What three parts make up a single nucleotide:

_____

9. What are the 4 bases that make up the rungs of the DNA ladder?

_____

10. What sugar is found in DNA? _____

11. What sugar is found in RNA? _____

12. How do the bases bond together? A bonds with _____ G bonds with _____

13. Why is RNA necessary to act as a messenger?

_____

14. Proteins are made where in the cell?

_____

15. How is RNA different from DNA? (list 3 things)

_____

_____

_____

16. The process of copying DNA is called _____

17. What is the shape of DNA? _____

18. Proteins are made from what subunits? _____

19. Three bases found on mRNA are called a _____

20. How do some cells become brain cells and others become skin cells, when the DNA in ALL the cells is exactly the same? In other words, if the instructions are exactly the same, how does one cell become a brain cell and another a skin cell?

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21. Why is DNA called the "Blueprint of Life"?

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